

Revolutionizing Fashion Retail: Virtual Try-on Empowered By Deep Learning

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Abstract—Online clothing purchasing has become a widespread practice for millions of individuals worldwide in the last several years. It has advanced significantly and is now a regular activity for many customers. The use of virtual try-on technology, which allows customers to view how particular things look on them prior to making a purchase, is what is facilitating this advancement. Shopping online is gaining more and more popular, especially for clothes. Nevertheless, the inability to physically try on clothing when shopping online limits customers' ability to visualize how a garment might look on them. The technology can produce high-fidelity try-on photos that maintain the features and general look of clothing items. Apart from the target garment region, the method further inserts the face, hair, lower garments, and legs to preserve model's personality characteristics. The two modules that comprise the system are the Try-On Module (TOM) and the Geometric Matching Module (GMM). In the Geometric Matching Module, in-store clothing items are warped to the intended image of a person with great precision using grid interval consistency loss and an occlusion management technique. The transformation is regularized by grid interval consistency loss to prevent distortion of clothing patterns. Additionally, an occlusion handling technique promotes appropriate warping even when target bodies are obscured by arms or hair. Then, the face, hair, lower clothing, and legs are added to the model's representation for the Try on Module in order to preserve personality characteristics other than the target clothing region.

Index Terms—Convolutional Neural Network(CNN), Deep Learning, U-Net, Virtual try-on(VITON)

I. INTRODUCTION

As virtual try-on technology has the ability to completely change the way consumers interact with products and make decisions about what to buy, it has become extremely important in the retail sector. It gives users the ability to electronically try on apparel and accessories, giving them a realistic idea of how the products might fit and seem without requiring them to engage physically. In the end, the ability to digitally try on things improves the overall customer experience and happiness by addressing common issues related to online buying, such as ambiguity about fit and style. Virtual try-on technology is therefore a useful tool in the e-commerce space, with the potential to minimize returns and maximize conversion rates for companies. Because it directly affects realism, accuracy, user experience, trust in purchase decisions, and the efficacy of personalized suggestions, high-quality data is essential to

the success of virtual try-on systems. Retailers hoping to improve their online shopping platforms with virtual try-on technologies must invest in high-quality data collecting and curation. To solve the virtual try-on challenge, one must try on a picture of a fabric on an image of the target person. Users can virtually experience wearing different outfits without having to physically change them according to this virtual try-on system. This virtual try-on technology not only helps customers save money by enabling them to virtually try on clothing, but it also improves the shopping experience for customers by changing the way they shop for clothing. Furthermore, this technique increases sales efficiency by assisting clients in making selections about what to buy and in determining whether or not they like a particular piece of clothing rapidly.

II. LITERATURE REVIEW

Virtual garment fitting makes use of image processing and deep learning algorithms. Using deep neural networks, these techniques fit the clothes according to the body parts and poses of the individual. A few virtual try-on techniques will be described in this section that follows.

The goal of Masahiro et al.'s unique virtual fitting method is to modify two-dimensional clothing images on models. Their three-dimensional models created from single-shot depth photos are used to carry this out. Next, the system uses a technique that superimposes clothing images that are acquired from an individual based to their body type.

First, the body image and the depth image, which were both taken at the same moment, are the input data that are being processed. The single-shot body forms are then estimated using the supplied data. By comparing input depth image with pre-trained depth images, these body form predictions ultimately provide an accurate three-dimensional body shape model for the user. The selection of apparel photos is the next processing phase. Using the 3D body form from a single shot body shape estimate as input, this processing phase searches the apparel database for comparable body shape models. A chosen image from the database of clothes that fits the various body shapes is the outcome of this method. The final stage of processing is transforming the photos of apparel. From this angle, the final overlaid image is produced by combining

the body image, depth image, three-dimensional body shape model estimates, and the chosen clothing image.

This system can generate extremely authentic models of apparel even when faced with physical and geometric limitations. It is also worth noting that this approach offers almost perfect control over both texture and view direction. On the other hand, the drawback is that it is extremely expensive. This is due to the fact that the procedure calls for the collection of data, annotation, computing power, and specialized tools with deep sensors. These enormous expenses frequently prevent the system from scaling to accommodate large numbers of users and clothing.

Next, using Conditional Generative Adversarial Networks (CGANs), Han et al. built VITON, which is able to seamlessly move the target clothing article of a product image to an accurate region of a clothed person in a two-dimensional image. Initially, they presented a human representation that was independent of clothes and consisted of multiple traits that were employed to depict distinct attributes of an individual. Given a reference image, this feature extracts the hair, body shape, face features, and even the pose.

Subsequently, they include the data from the clothing-agnostic representation into the multi-task encoder-decoder generator in the following section. To create a coarse, artificially dressed human, they employed a multi-task encoder-decoder generator. This character is designed to be in the same position, to be dressed in the same target attire, and to have a mask that matches the clothing region. Subsequently, the target clothing picture and clothing mask are used in shape and context matching to estimate the thin-plate spline (TPS) transformation, which in turn produces a warped image of the clothes. Additionally, they leverage a refinement network that has been extensively trained with the goal of understanding how to correctly composite a distorted article of clothing to a coarse image. This is done to ensure that the target object transfers smoothly and naturally while maintaining a rather intricate visual pattern. VITON's primary function, which primarily uses RGB photos, is to convert an article of clothing in a product image to a human. The drawback is that VITON struggles to preserve the unique qualities of fabric, like texture and branding. Han and colleagues, (2018).

The architecture structure of CP-VTON is also the foundation for CP-VTON+. The architecture is improved in a number of locations. First, during the GMM stage, all of the VITON dataset is relabelled as the naked chest area, the neck is incorrectly classified as the backdrop, and occasionally hair occlusion will cause the body's shape to be incorrectly deformed. Second, they utilize clothing masks in place of the colored texture from try-on clothes, since it hinders the matching process. The addition of regularization for the TPS parameters is the final modification made to the GMM network. The reason for the regularization is that the technologies now in use indicate that clothing that has warped is frequently significantly distorted.

$$L_{GMM}^{CP-VTON+} = \lambda_1 \cdot L1(C_{warped}, I_{C1}) + \lambda_{reg} \cdot L_{reg} \quad (1)$$

$$L_{reg}(G_x, G_y) = \sum_{i=-1,1} \sum_x \sum_y |G_x(x+i, y) - G_x(x, y)| \quad (2)$$

$$\sum_{j=-1,1} \sum_x \sum_y |G_x(x, y+j) - G_x(x, y)| \quad (3)$$

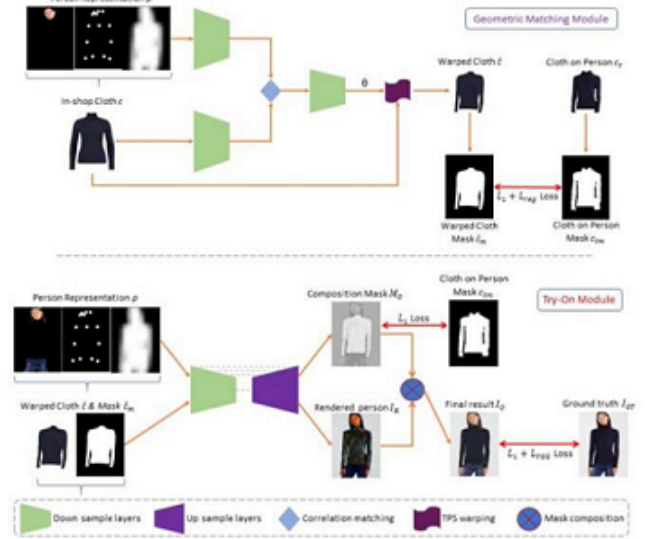


Fig. 1. Overview of CP-VTON+ pipeline (Matiur et al., 2020)

The face, lower garments, hair, and legs were added to the TOM stage's human representation input. In contrast to the target garment area, they do this in order to preserve the other human feature. In order to prevent the TOM from mistaking the white clothing area for the background of the in-store clothing image, they additionally included the binary mask of twisted clothes to the TOM network input (Matiur et al., 2020).

III. PROPOSED METHOD

In addition to outlining the recommended pipeline design, this part offers a thorough breakdown of every component and justification for the models chosen. The next subsection contains information on the frameworks and tools used to create this process. This system was modeled after CP-VTON+, which was created by Matiur et al. (2020). The three subtasks that comprise the outline-course-fine technique used by the system are the Geometric Matching Module (GMM), the Try-on Module (TOM), and model representation. GMM's main objective is to align the desired fabric with the user's body in order to make a warped cloth. GMM uses grid interval consistency and occlusion handling to precisely mold the necessary cloth around the model. After that, the warp cloth will be delivered to TOM, which will process it and produce the last try-on photo of the model dressed in the selected fabric. TOM features a synthesizing discriminator to further improve the quality of the images it produces.

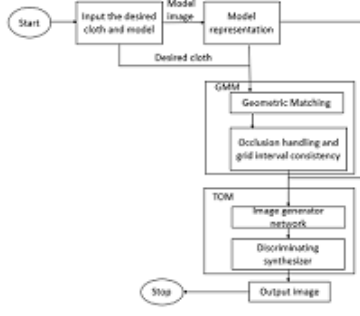


Fig. 2. Overview of the proposed method)

The study "Convolutional neural network architecture for geometric matching", released by Rocco et al. in 2017, serves as the foundation for the majority of the system's Geometric Matching Module. In this study, a unique convolutional neural network architecture is used to estimate the parameter of a geometric transformation between two input pictures. Using a local descriptor, the two input photographs are extracted in the first stage. To generate a list of preliminary correspondences, descriptors from various photos are matched in the second stage. They present an architecture that seeks to resemble the same procedure, which involves a convolutional layer that receives input images I_A and I_B .

Convolutional layers are used to extract I_A and I_B feature maps, which are comparable to dense local descriptors. The next step is to create a preliminary correspondence map(f_{AB}) by matching the feature maps across pictures. Next, a regression network outputs the geometric model's parameters, θ , in a reliable way. The geometry estimation tasks may be trainable end-to-end using this Geometric Matching Module (Rocco et al., 2017).

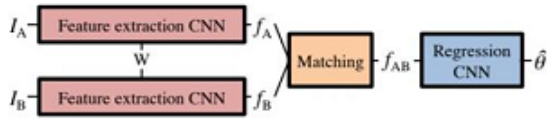


Fig. 3. GMM architecture)

In this system, a new GMM is applied to convert the intended clothing into distorted garments. There are four sections in the new GMM. Two networks are introduced and the high-level properties of human representation are extracted in the first phase. In the second section, a correlation layer is inserted and two features are combined into a single tensor. The regression network uses a single tensor from the correlation layer to predict the spatial transformation parameters in the third segment. The final step is utilizing a TPS to warp a picture into the output.

The primary intent of the try-on module is to locate the intended receiver with the wrap cloths. The try-on module combines two techniques. Preserving the characteristics of the folded fabric—which will be directly applied to the recipient's image—is the first stage. Nevertheless, this has the disadvantage of occulting some desired body parts and giving the cloth's edge an unusual aspect. Another approach is U-net, which Olaf et al. introduced in 2015.

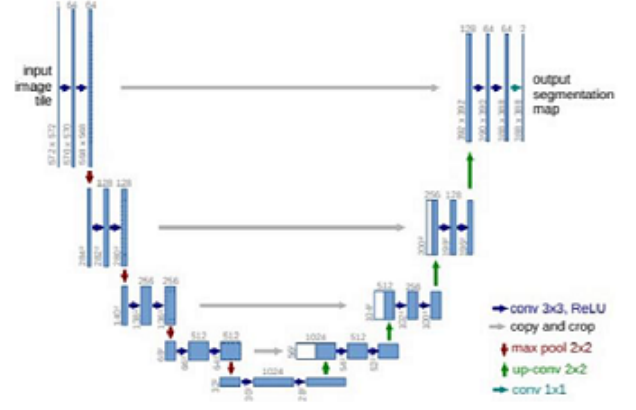


Fig. 4. U-net architecture

U-net's design is made up of two paths: the left path, which contracts, and the right path, which expands. An up sampled feature map is followed by a 2x2 convolution (sometimes called a "up-convolution") that halves the number of feature channels for the expanded path. After that, there is a concatenation with the contracting path's equally cropped feature map, then two 3x3 convolutions, each of which is followed by a rectified linear unit. On the other hand, the contracting path entails down sampling by repeatedly applying two 3x3 convolutions, each of which is followed by a rectified linear unit (ReLU) and a 2x2 max pooling operation with stride 2. A 1x1 convolution at the bottom final layer maps each 64-component feature vector to the required number of classes. Thus, there are 23 convolutional layers in all. One disadvantage of using U-net, though, is that it cannot explicitly modify space; even a small misalignment can result in a hazy U-net rendered output (Olaf et al., 2015). As a result, try on module employs technique to produce a visually appealing output. After that, it predicted the rendering of a composition mask and a person image using U-net. Finally, a composition mask will be applied to the rendered figure and the warp cloth to create the person wearing the cloth.

IV. DATASET

The VTON dataset was obtained to be used (Han et al., 2018). It has about 19,000 image pairs of women wearing tops and front views. 16,253 of the 19,000 cloths can be used in pairs, and these are divided into training and testing sets. There are 14,221 image pairs in the training set and 2,032

image pairs in the testing set. The dataset also includes person parse, cloth mask, and human posture key point. In order to test, the testing set is also rearranged into unpaired pairs.



Fig. 5. Cloth images



Fig. 6. Training image of the cloth mask

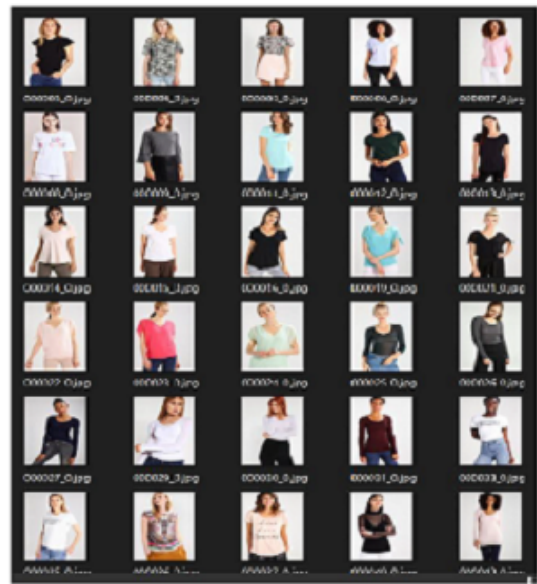


Fig. 7. Image of model wearing the cloth



Fig. 8. Training image of the person parse

V. TESTING

There was a total of 2,032 image pairs used for testing; each pair included a front-view photo of a woman and an image of top garments with a resolution of 256 X 192. This dataset is used by many of the older virtual try-on systems to test their systems. The system uses the model's image and the clothing image as input to create the display image.



Fig. 9. First Test Case



Fig. 10. Second Test Case

VI. RESULTS

The suggested approach produces a very genuine try-on outcome. In addition, the system can deal with the misalignment of the subject and the fabric, maintaining the features of the target cloth (such as the face, lower body, and hair) and the non-target cloth (such as the clear body parts). Apart from that, it can also simulate connection in between the model and the clothing, which helps the proposed system to prevent distortion in the logo, embroidery, and fabric characteristics. Additionally, human features like the hand, face, and lower garments are retained.

However, there are still some issues with the system, particularly when there is an arm in front of the cloth, making it unable to fully resolve the occlusion problem.

A cloth and an image of a model are the inputs utilized to create the try-on results for a virtual try-on during the test.

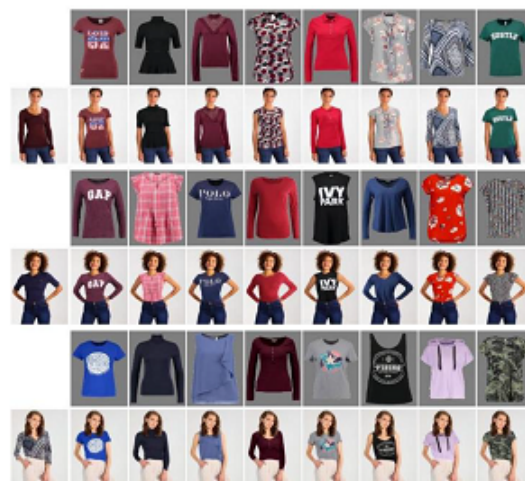


Fig. 11. Successful cases



Fig. 12. Unsuccessful Cases

Consequently, the measure is assessed using the FID. The quality of the outcomes increases with a lower FID score. This system has the lowest Frechet Inception Distance based on the calculation.

The table below shows that the system's FID score is lower than that of other systems indicating that the image that it produces is precise.

METHOD	FID
CP-VTON	24.43
CP-VTON	21.08
PROPOSED SYSTEM	10.09

The below plotted graph shows the time taken by each process.

VII. CONCLUSION AND FUTURE WORK

When it comes to e-commerce, online clothing holds the largest market share and is expanding at an incredibly rapid

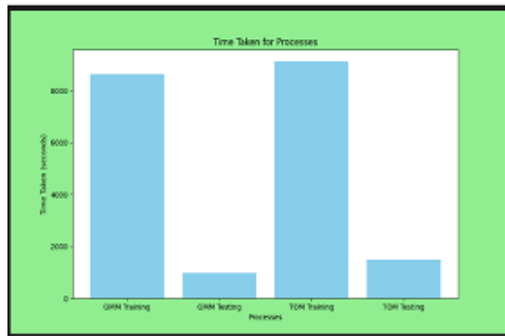


Fig. 13. Time taken by each process

rate. This is due to the convenience of shopping for clothing online while lounging in one's own home. However, the drawback is that because online purchasing does not permit putting on clothes in person, customers are limited in their ability to visualize how a garment would seem on them. The majority of existing methods create pictures using a coarse-to-fine network in order to create realistic images. The human parsing characteristic is completely eliminated in the current technology, resulting in an unrealistic and hazy appearance.

The system's benefit is that it may preserve the apparel product's features, such as the texture, design, and their brand, without introducing distortion or obvious artifacts. In addition to the target garment region, the technology is also able to maintain other human traits, such as the face, lower body, legs, and hair. In addition, the system has the capability to handle hair occlusion. Its inability to convert the input image into the target person's 3D posture is a drawback, too. When a human image is inputted with the hand blocking most of the body, the system cannot perform occlusion and will produce distorted images. In order to tackle occlusion in virtual try-on, we can investigate using the reference person's 3D structure as a future upgrade for this system.

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